

EXPERIMENT 2:

Implement a basic finite state automaton that recognizes a specific language or pattern. In this example, we'll create a simple automaton to match strings ending with 'ab' using python.

AIM:

To implement a basic Finite State Automaton (FSA) that recognizes strings ending with the pattern "ab" using Python.

PROGRAM:

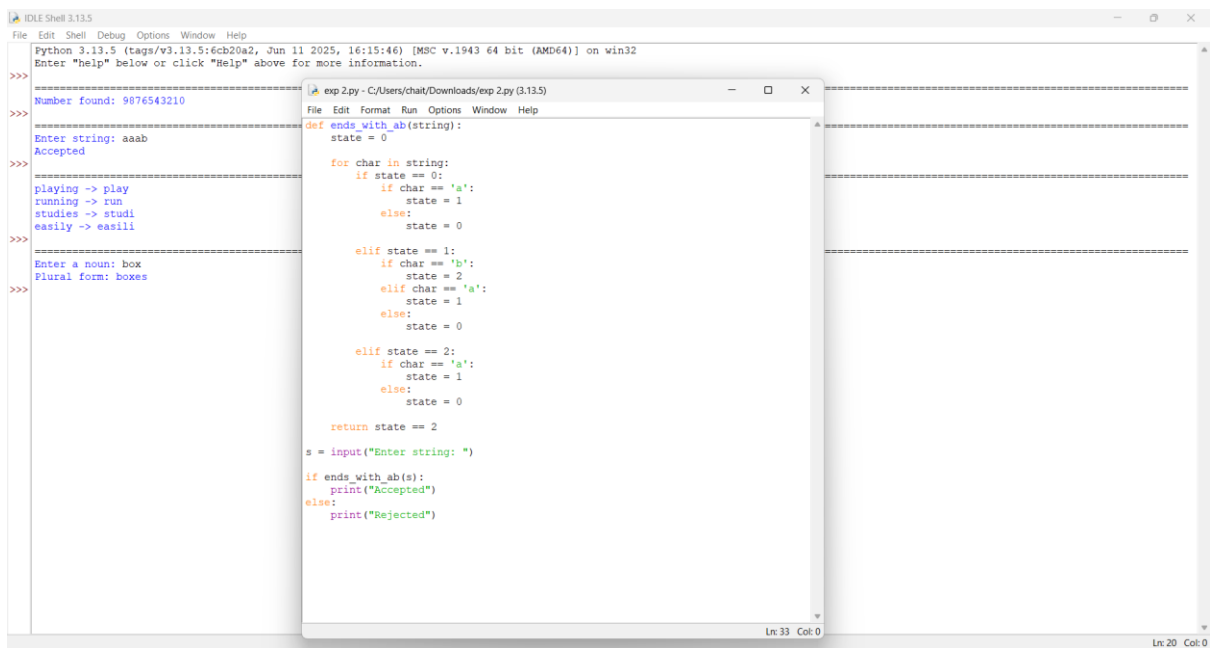
```
def ends_with_ab(string):  
    state = 0  
    for char in string:  
        if state == 0:  
            if char == 'a':  
                state = 1  
            else:  
                state = 0  
        elif state == 1:  
            if char == 'b':  
                state = 2  
            elif char == 'a':  
                state = 1  
            else:  
                state = 0  
        elif state == 2:  
            if char == 'a':  
                state = 1  
            else:  
                state = 0  
    return state == 2  
s = input("Enter string: ")
```

```
if ends_with_ab(s):  
    print("Accepted")
```

```
else:
```

```
    print("Rejected")
```

OUTPUT:



The screenshot shows a Python IDE with two windows. The background window displays the output of a program, and the foreground window shows the source code.

Background Window Output:

```
>>>  
Number found: 9876543210  
>>>  
Enter string: aaab  
Accepted  
>>>  
=====  
playing -> play  
running -> run  
studies -> studi  
easily -> easili  
>>>  
=====  
Enter a noun: box  
Plural form: boxes  
>>>
```

Foreground Window Code (exp2.py):

```
def ends_with_ab(string):  
    state = 0  
  
    for char in string:  
        if state == 0:  
            if char == 'a':  
                state = 1  
            else:  
                state = 0  
  
        elif state == 1:  
            if char == 'b':  
                state = 2  
            elif char == 'a':  
                state = 1  
            else:  
                state = 0  
  
        elif state == 2:  
            if char == 'a':  
                state = 1  
            else:  
                state = 0  
  
    return state == 2  
  
s = input("Enter string: ")  
  
if ends_with_ab(s):  
    print("Accepted")  
else:  
    print("Rejected")
```